

Overview

ET

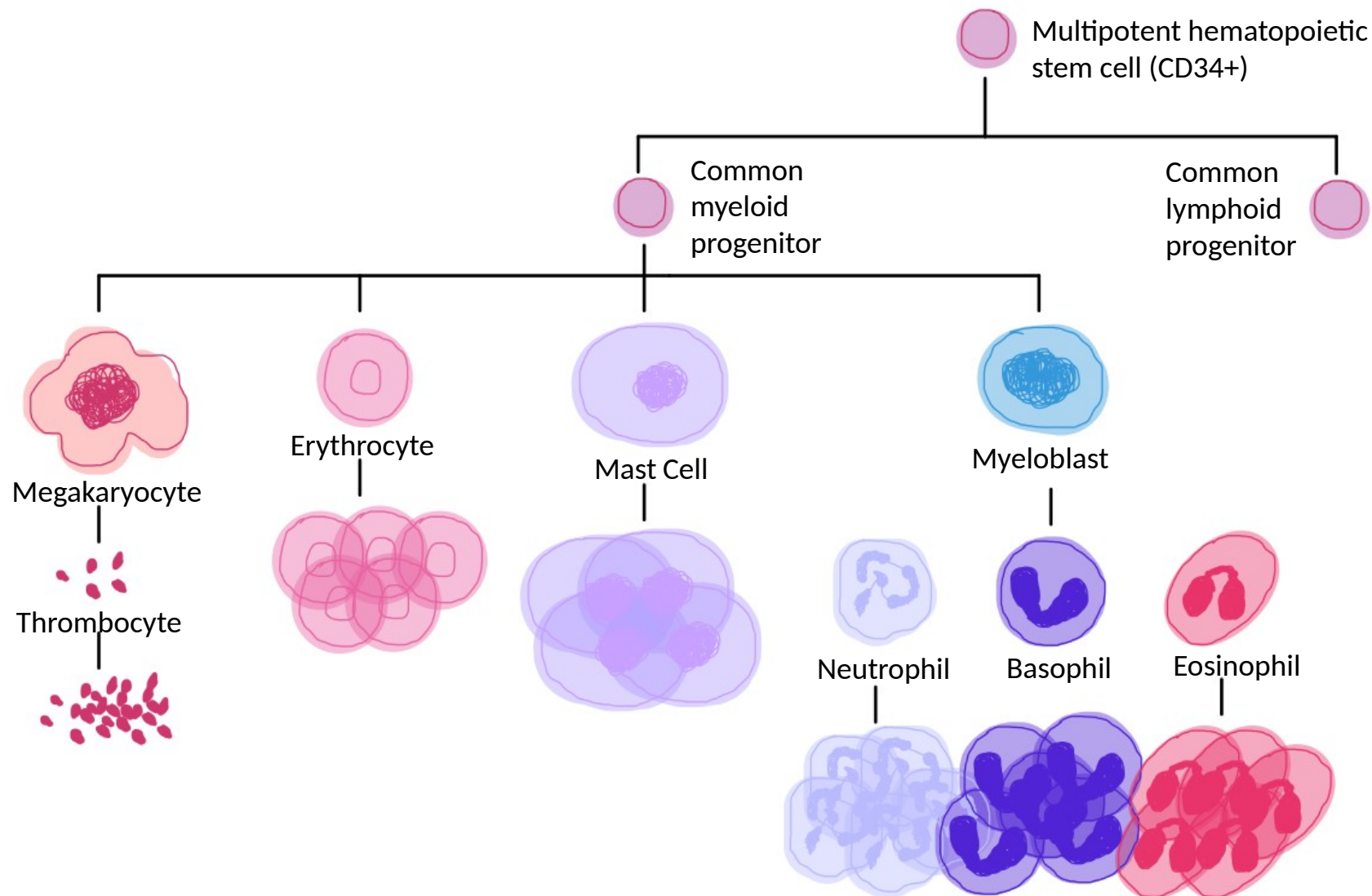
PV

PMF

CML

Cases

Myeloproliferative Disorders



Overview

ET

PV

PMF

CML

Cases

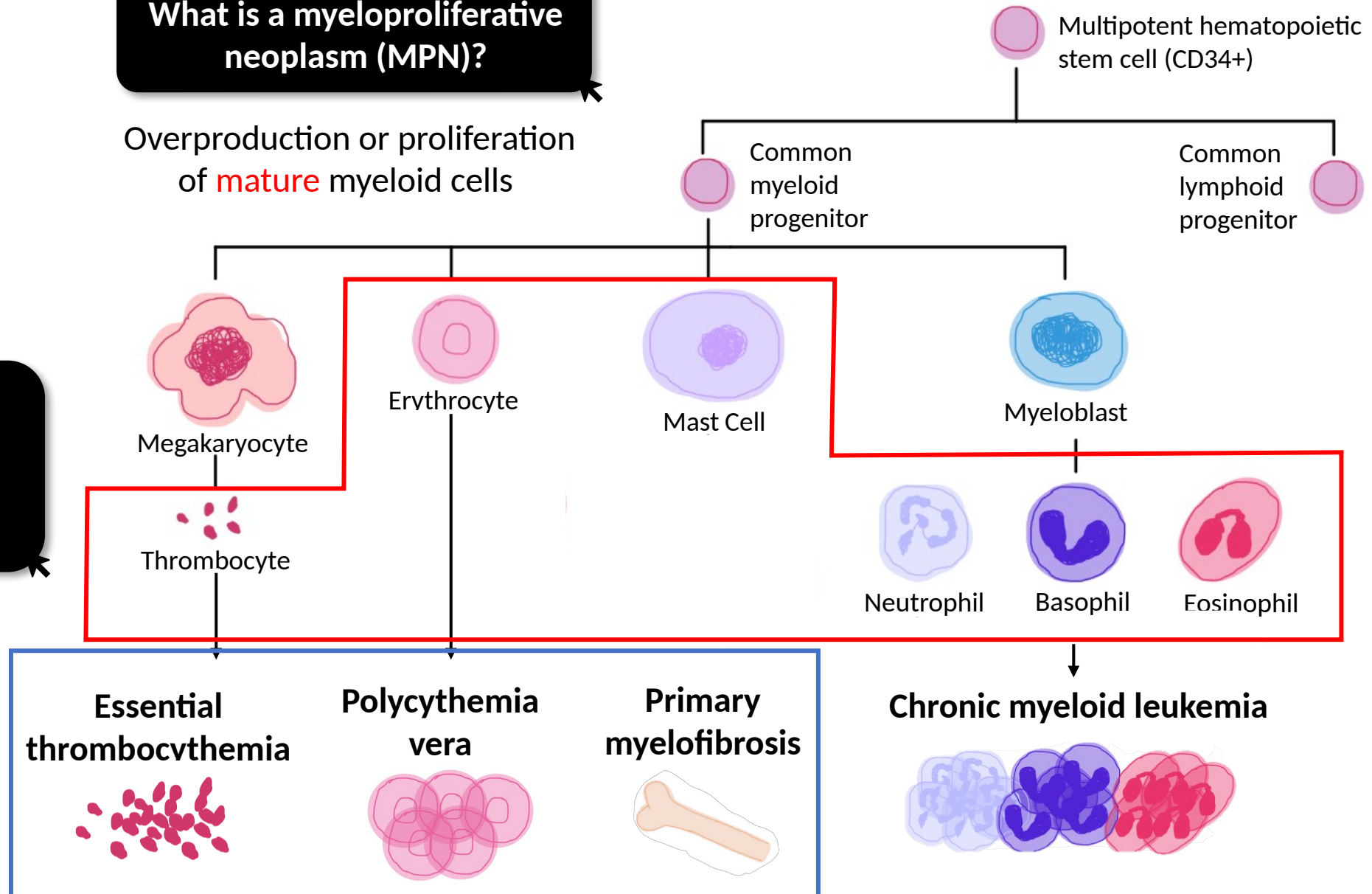
What are the most common MPNs?

Associated with 3 driver mutations:
JAK2, CALR, MPL

Overview

What is a myeloproliferative neoplasm (MPN)?

Overproduction or proliferation of **mature** myeloid cells



Essential Thrombocythemia



What is ET?

- Increased clonal platelet production
- Associated with 3 driver mutations: JAK2, CALR, MPL

Signs/Symptoms?

Asymptomatic, vasomotor symptoms, thrombosis, bleeding

Differential?

Reactive Thrombocytosis

- Iron-deficiency anemia
- Infection
- Inflammation
- Post-splenectomy

Primary Thrombocytosis

- ET
- PV
- PMF
- CML

Treatment of ET

Goal is to prevent thrombosis!

↑ platelet ≠ ↑ thrombosis

Don't treat reactive thrombocytosis!

Low risk

< 60 yo
No history of thrombosis

Observation
Low dose aspirin

High risk

> 60 yo OR
History of thrombosis

Venous thrombosis

Anticoagulation +
Cytoreductive agent

Arterial thrombosis

Low dose aspirin +
Cytoreductive agent

- Overview
- ET
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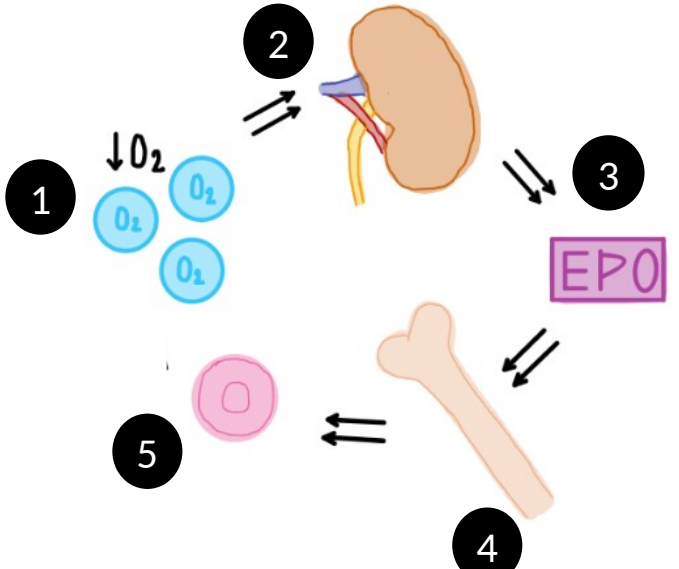
What is PV?

- ↑ production of ≥1 cell lines
- Hallmark: ↑ in RBC mass independent of EPO level (↑ plt and WBC also common)
- >95% have JAK2 V617F mutation

Signs/Symptoms?

- ↑ 25% RBC mass
- Aquagenic pruritus
- Headache
- Thrombosis
- TIAs
- Bleeding

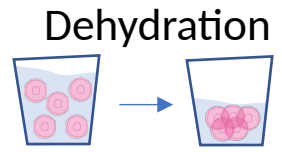
Physiology of erythrocytosis



Polycythemia Vera

Click until cursors disappear

Primary problem



Examples

- Diuretics
- Vomiting
- Diarrhea

EPO level

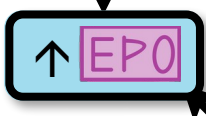
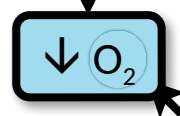
↔

Differential?

Relative

Secondary

PV



- COPD, OSA
- High elevation
- RAS

- HCC
- RCC
- Pheo

↑↑

↔ / ↓

Treatment of JAK2 + PV

Goal is to prevent complications (e.g., thrombosis)!

Low risk

< 60 yo
No history of thrombosis

Phlebotomy (goal hct <45%)
Low dose aspirin

High risk

> 60 yo OR
History of thrombosis

Phlebotomy (goal hct <45%)
Low dose aspirin +
Cytoreductive agent (hydroxyurea)

Overview

ET

PV

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CML

Cases

Primary myelofibrosis

What is PMF?

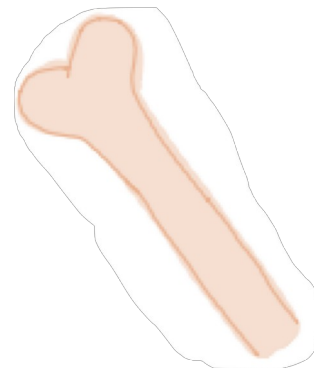
- Bone marrow replaced by connective tissue (fibrosis)
- Can develop from ET or PV
- Associated with JAK2 mutation

Signs/Symptoms?

- Hepatosplenomegaly
- Bone pain
- Anemia
- Bleeding
- Infections

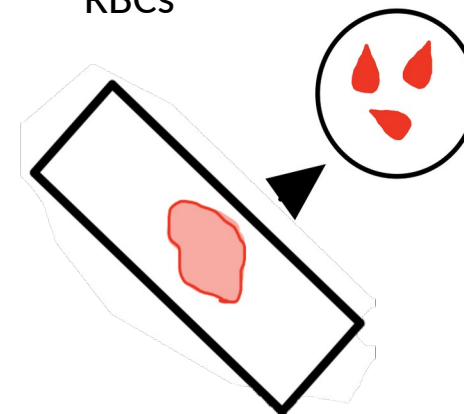
Bone Marrow Biopsy

- Fibrosis
- “Dry” tap may be seen
- Megakaryocyte hyperplasia



Peripheral smear

- Teardrop -shaped RBCs



Treatment

Dependent on prognosis

Good prognosis

Active surveillance
Supportive care for cytopenias (transfusions)

Bad prognosis

Consider hematopoietic stem cell transplant
in addition to supportive care

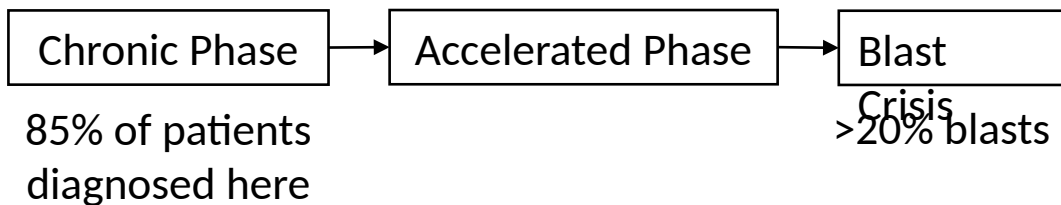
Chronic Myeloid Leukemia



What is CML?

- ↑ production of partially mature granulocytes
- **BCR-ABL1** fusion gene (**Philadelphia Chromosome**) → ↑ tyrosine kinase activity
- Cytochemically abnormal (low LAP score)

Natural History



Differential?

Leukemoid Reaction

- Infection
- Stress response
- Chronic inflammation
- Bone marrow stimulation
- Medication-induced
- Splenectomy

Malignant etiologies

Typically malignant if WBC > 100k

- | | |
|-------|-------|
| • CML | • AML |
| • CLL | • ALL |

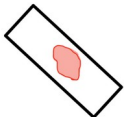
How to further differentiate

Treatment

- First line treatment: tyrosine kinase inhibitors (TKI)
- Blast crisis CML may require induction chemotherapy (like acute leukemia)

- Overview
- ET
- PV
- PMF
- CML**
- Leukemias**
- Cases

Overview of Leukemias

		AML	CML	ALL	CLL
Click until cursor disappears					
Peripheral smear		Myeloblasts Auer rods	Eosinophilia/ basophilia Median WBC 100k	Lymphoblasts	Lymphocytosis Smudge cells
Bone marrow		>20% myeloblasts	Granulocytic hyperplasia Typically <20% myeloblasts	>20% lymphoblasts	Lymphocytic hyperplasia
Demographic		Adults > Children	Median age 50	Children > Adults	Median age 70
Unique features		DIC (especially in APML)	T(9;22) BCR-ABL1 fusion	- CNS infiltration (needs LP prior to treatment) - Can infiltrate lymph nodes	- Richter transformation - Autoimmune cytopenias - Can infiltrate lymph nodes

Overview

ET

PCV

PMF

CML

Cases

Case 1

53 yo F presents with fatigue.

Vitals: T 37.6C, HR 76, BP 118/77. SaO₂ 99%.

CBC shows:

10.7 17.4 423
51.6

What additional history do you want?

CBC shows **erythrocytosis**. Evaluate for symptoms of PV (headache, aquagenic pruritus, thrombosis) and identify RF for 2° polycythemia (smoking, cancer).

What test would you order next?

EPO level. Although a normal EPO level does not rule out PV, a ↓ EPO level raises clinical suspicion for PV.

...She reports headache, itching after showering. She denies smoking or history of cancer. EPO level is low.

What is the next step in evaluation?

JAK2 mutation testing.

She is diagnosed with PV. What are the next steps?

- Start low dose ASA for thrombosis ppx
- Hematologist/oncologist referral

Case 2

45 yo presents to clinic to establish care. A routine CBC shows:

29.3 11.4 134
35.3

What is your ddx for the leukocytosis?

Infection, chronic inflammatory process, med-induced (e.g. steroids or drug reaction), leukemoid reaction (surgery, GI bleed), tobacco use, hematologic malignancy

What is the next step in evaluation?

Peripheral smear (or differential).

Smear shows basophilia + eosinophilia and promyelocytes. What is the next step in evaluation?

Bone marrow biopsy. *Explanation of why this is abnormal.*

...Bone marrow biopsy shows granulocyte hyperplasia with 10% myeloblasts. Cytogenetics reveal t(9;22)

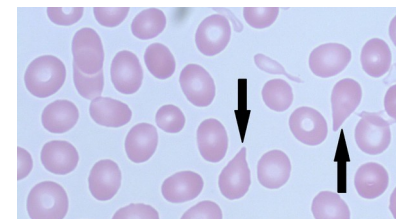
What is the diagnosis?

Chronic myeloid leukemia

Case 3

67 yo presents with 1 month of malaise and general weakness. Initial labs show:

3.9 10.6 98
34.1



What does the smear show?

Teardrop cells

What is the next step in evaluation?

Bone marrow biopsy

...Initial biopsy attempt was a dry tap. Pathology shows reticular fibrosis and megakaryocyte hyperplasia.

What is the diagnosis?

Primary myelofibrosis

What are next steps in management?

- Determine prognosis for HSCT
- Symptom management with transfusions for cytopenias

	ET	PV	PMF	CML
What is it?				
Signs & Symptoms				
Differential				
Treatment				

	AML	CML	ALL	CLL
Peripheral smear				
Bone marrow				
Demographic				
Clinical features				

Myeloproliferative Neoplasms - 2022-2025 Update

What's changed since this talk (May 2022)

Diagnosis still rests on WHO criteria: PV needs Hgb >16.5 (M) / >16.0 (F) -- OR Hct >49% (M) / >48% (F) -- plus marrow panmyelosis and a JAK2 mutation (low EPO is the minor criterion). The deck's Hgb thresholds remain correct; WHO 2022 simply adds the Hct option.

Polycythemia vera: ropeginterferon alfa-2b (Besremi) was FDA-approved (Nov 2021) as the first PV-specific interferon and is now an NCCN preferred first-line cytoreductive agent -- the named product behind the deck's generic "pegylated interferon."

Myelofibrosis: there are now FOUR JAK inhibitors, chosen by blood counts -- ruxolitinib and fedratinib (baseline), pacritinib (Vonjo; FDA Mar 2022) for severe thrombocytopenia (platelets <50k), and momelotinib (Ojjaara; FDA Sep 2023) for MF with anemia. The deck names only ruxolitinib.

CML: asciminib (Scemblix), a STAMP inhibitor that binds the ABL myristoyl pocket, was FDA-approved (Oct 2021) for ≥ 2 prior TKIs or the T315I mutation -- a distinct mechanism beyond the deck's "next-generation TKIs," since expanded toward first-line use.

Standing teaching unchanged: the four MPNs, the JAK2/CALR/MPL driver mutations, the erythrocytosis/thrombocytosis/leukocytosis differentials, EPO physiology, risk-stratified ET/PV treatment, and the AML/CML/ALL/CLL comparison all remain current.

Myeloproliferative Neoplasms

Take-Home Points

Four main MPNs -- essential thrombocythemia, polycythemia vera, primary myelofibrosis, and chronic myeloid leukemia -- arise from clonal overproduction of MATURE myeloid cells (vs MDS = abnormal maturation). ET/PV/PMF share the JAK2/CALR/MPL drivers; CML is defined by BCR-ABL1.

A low EPO level helps separate polycythemia vera (suppressed EPO, >95% JAK2 V617F) from secondary erythrocytosis (elevated EPO: hypoxia, EPO-secreting tumor) and relative erythrocytosis (normal EPO, hemoconcentration).

Treatment of ET and PV is risk-stratified by age >60 and prior thrombosis: aspirin/phlebotomy (PV goal Hct <45%) for low risk, add a cytoreductive agent for high risk. Platelet count magnitude does NOT drive thrombosis risk -- don't treat reactive thrombocytosis.

PMF management is prognosis-driven (HSCT is the only cure; JAK inhibitors for symptoms). CML is treated with TKIs and is now highly effective -- most patients reach a near-normal life expectancy.

Myeloproliferative Neoplasms

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Adapted from TeachIM.org -- "Myeloproliferative Neoplasms" (Hematology). Authors: Stephanie Lakritz, MD; Christopher Geiger, MD; Brandon McMahon, MD. Section Editor: Christopher Geiger, MD; Executive Editor: Yilin Zhang, MD. Published May 2022.

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Optimization (June 2026): Guideline currency verified. One deck typo fixed surgically -- "aquagenic pruritis" -> "aquagenic pruritus" (slide 8, Case 1); all 9 original animated/click-to-reveal slides and BioRender graphics preserved. Three slides appended: 2022-25 update, take-home, and this attribution. Newer context added (all source-cited): WHO 2022 PV criteria (Hct option); ropeginterferon alfa-2b (Besremi) for PV; pacritinib and momelotinib joining the MF JAK-inhibitor class; asciminib (Scemblix) STAMP inhibitor for CML.

Companion files: facilitator guide (MPN-final.docx) and learner worksheet (MPN-worksheet.pdf), both copied into the lesson folder and linked from the HTML launcher.